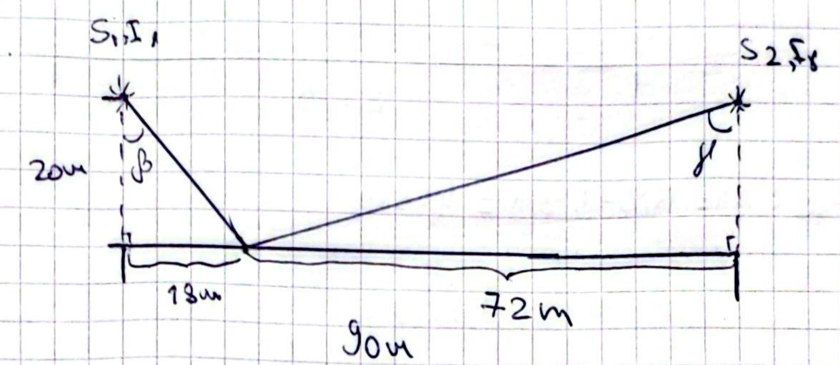
Murjabbar Badalov | Homework 3 | 285269

Problem 5:

2 uniformly radiating sources [90 m] apart, luminous intensity [15000 cd]
Located [20 m] height, is located at the projection of the source line
on the floo at 3/12 m distance from one of the sources.

$L = 90m$
$I_\gamma = 15000cd$
$H = 20m$
$X = 3y$
$Y = 12y$



Let's take distance from source 1 projection, for finding the point on the surface

$X = 3y$ $\quad$ $3y + 12y = 90$ $\quad$ $3 \cdot 6 = 18m$
$Y = 12y$ $\quad$ $15y = 90$ $\quad$ $12 \cdot 6 = 72m$ } from Source 1 projection
$y = 6$

We have to use cos law for calculating Illuminance for each source:

$$E = \frac{I_\gamma}{r^2} \cdot \cos\beta$$

$$E_1 = \frac{15000}{20^2} \cdot \frac{20}{\sqrt{20^2 + 18^2}} = \frac{15000}{20 \cdot 26{,}907} = 27{,}87\ lx$$

$$E_2 = \frac{15000}{20^2} \cdot \frac{20}{\sqrt{72^2 + 20^2}} = 10{,}03\ lx$$

$$\boxed{E_{total} = E_1 + E_2 = 37{,}9\ lx\ total}$$

[Problem 5 Continue with other way]

$$E = \frac{I_\gamma}{r^2} \cos\beta$$

in this way: r = total distance from source to point

$$E_1 = \frac{15000}{\left(\sqrt{20^2+18^2}\right)^2} \cdot \frac{20}{\sqrt{20^2+18^2}} = 15,4\ lx$$

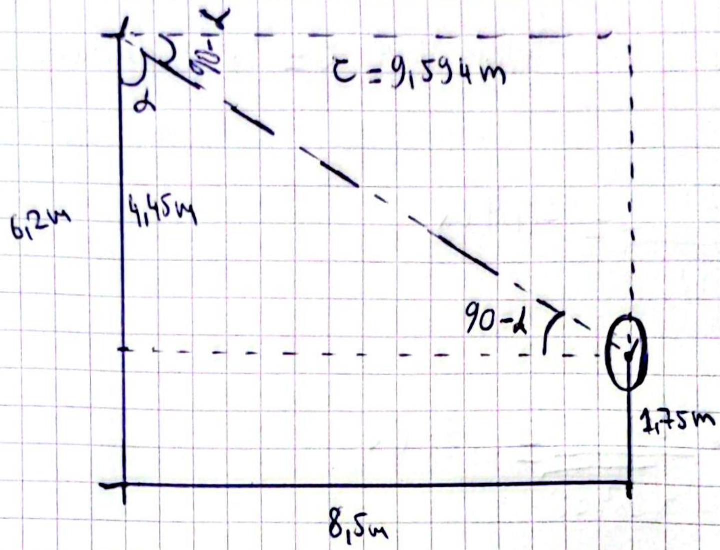
$$E_2 = \frac{15000}{\left(\sqrt{20^2+72^2}\right)^2} \cdot \frac{20}{\sqrt{20^2+72^2}} = 5,55\ lx$$

$$E_{total} = E_1 + E_2 = 20,95\ lx$$

[Problem 6:]

$I_\alpha = I_0 \cos(\alpha)$

$I_0 = 1100$



H difference: $\Delta H = 6{,}2 - 1{,}75 = 4{,}45\ m$

Distance from source to sign center: $r = \sqrt{8{,}5^2 + 4{,}45^2} = 9{,}594\ m$

$\cos\alpha = \frac{4{,}45}{9{,}594} = 0{,}464$ then $I_\alpha = 1100 \cdot 0{,}464 = 510{,}4\ cd$

Sign surface is vertical, for finding normal illuminance on its center:

$$E_n = \frac{I_\alpha}{r^2} \cdot \cos(90-\alpha) = \frac{I_\alpha}{r^2} \cdot \sin\alpha =$$

$$\left[= \frac{510{,}4}{9{,}594^2} \cdot \frac{8{,}5}{9{,}594} = 4{,}91\ lx \right]$$